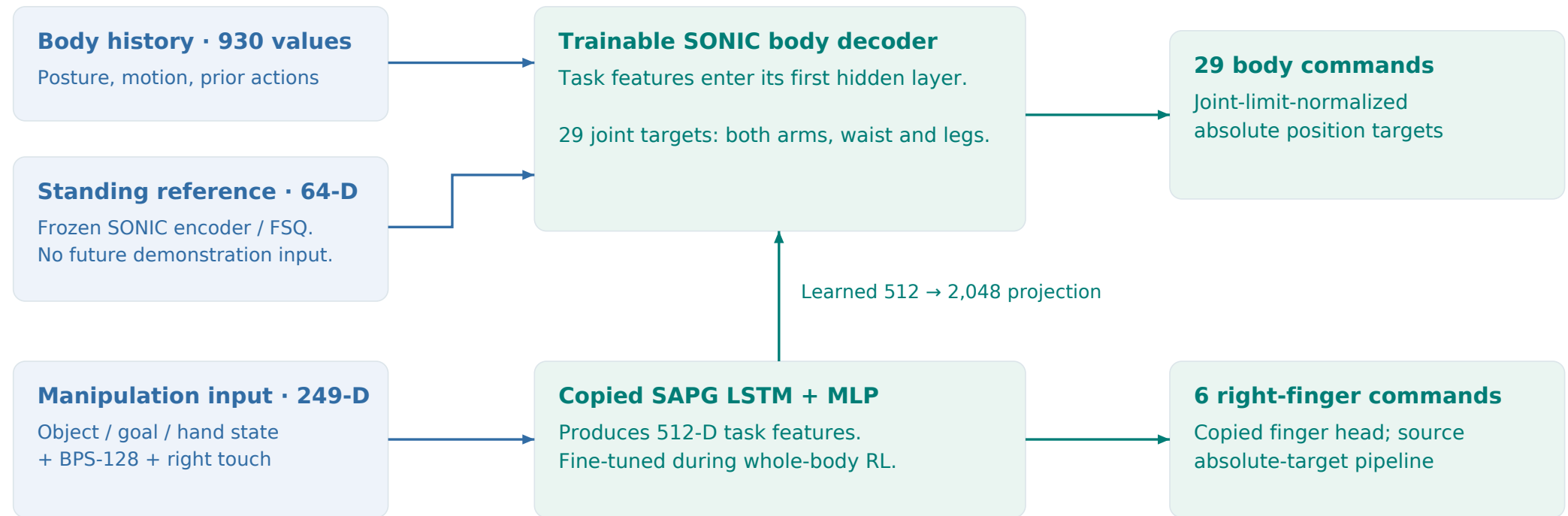


One policy for manipulation and the whole body

Implemented route 2: reuse the trained SAPG skill and fine-tune SONIC itself. This supersedes the frozen-decoder / latent-adaptor proposal.

STATUS: RL implemented and run to 1.233B transitions; stopped by CUDA out of memory.

ACTUAL CONTROL GRAPH



One motor owner: a single actor emits 35 actions. The old seven-arm SAPG output is not another controller; it is used only as an offline teaching target. The copied critic gains body inputs. During RL, decoder, task projection, copied task features and finger head can all learn.

Manipulation is connected to leg control, not to a footstep planner. The robot starts beside the table with a standing reference. It can shift weight or step, but purposeful walking-to-grasp and disturbance recovery have not been validated.

Reuse first, then learn with full-body physics

The completed BPS + touch teacher is reused. Offline imitation initializes the changed body-action interface; online SAPG then trains the combined controller.

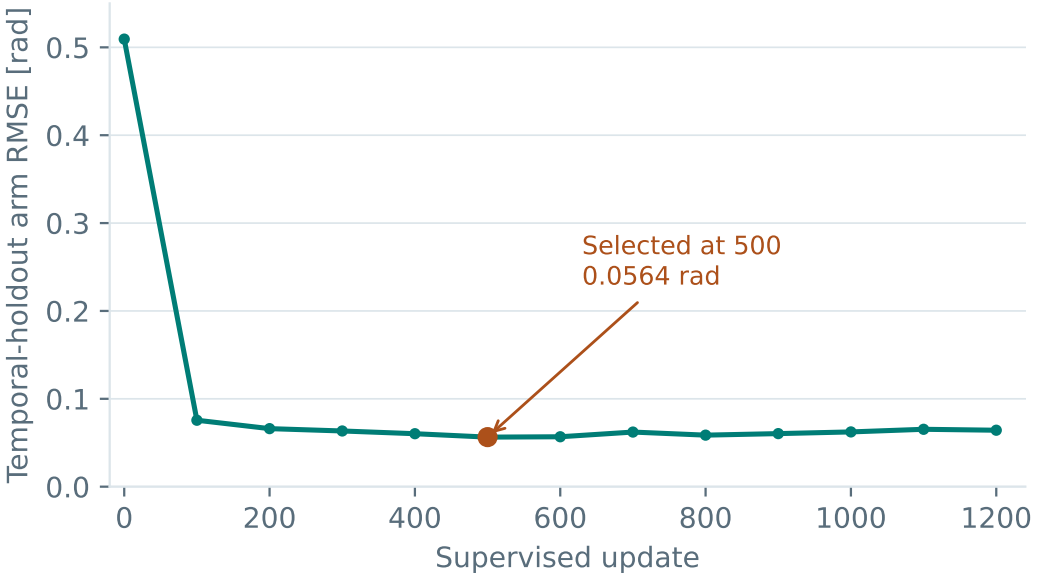


WHAT THE BOOTSTRAP ACTUALLY SAW

Four 30-second teacher clips at 60 Hz: brush, eraser, hammer and spatula; 7,200 captured samples. The right-arm motion is lifted into a nominal standing body **kinematically**, not replayed through full-body contact physics.

5,038 training / 1,862 validation rows, split within episodes with a 50-step gap. After recurrent burn-in, 1,668 holdout rows are scored. These are **not unseen objects**.

Separate real standing-physics rehearsal: 693 training / 247 validation rows at 50 Hz. All body histories retain SONIC’s 20 ms spacing.



Selected arm-target RMSE: 0.0564 rad (initial: 0.5094 rad). Selection combines arm, finger, other-body and standing errors. This measures target imitation—not grasping success.

No online imitation loss is currently added. Teacher/standing supervision is offline only; whole-body RL uses the original manipulation reward. This differs from the old proposal’s “SAPG + teacher loss” wording.

Module	Bootstrap	Online RL
SONIC body / task projection	Train	Train
SAPG task / finger features	Frozen	Train
Reference encoder / FSQ	Frozen	Frozen

Preserve the task; change the body interface

The environment subclasses the actual Play2Perfect → BPS → touch task. Startup checks reject unintended reward, goal, observation, timing or object-bank changes.

Simulation component	Implemented setup
Floating G1 + two Revo2	51 joints / 68 rigid bodies; gravity on
Policy / physics / touch	60 Hz / 120 Hz / 70 Hz
Controlled outputs	29 body + 6 right hand = 35
Left hand	Physical joints present, held neutral
Object bank	1,200 procedural tools; same bank SHA
Right-hand drives	Source PD + five target followers
Fabrics / PCA / arm torques	All disabled in this run

OBSERVATION ACCOUNTING

Actor: 92 task + 132 BPS + 25 touch = 249.
Body extension: 930 history + 64 standing code = 994.
Total: actor 1,243; critic 1,265, before SAPG coefficient conditioning.

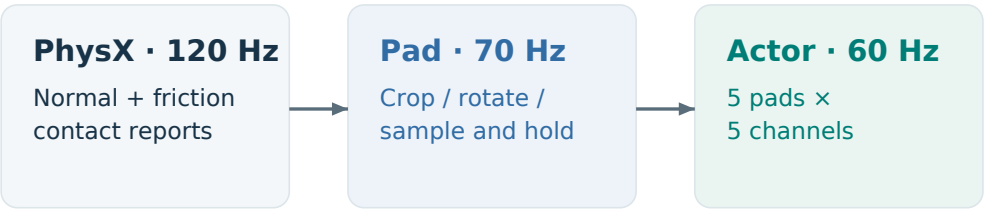
Body targets are absolute; source finger smoothing, target coupling and action delay are preserved. Original arm-delta calculations do not write the body joints.

SHAPE AND TOUCH REACH THE POLICY

BPS describes shape; keypoints describe the goal

128 distances to normalized surface samples + centroid (3) + radius (1) = 132 features. The original four goal-alignment keypoints are retained separately.

$$d_i = \min_{p \in S} \|b_i - p\|_2, \quad i = 1, \dots, 128$$



Per pad: normal force, two shear components, validity and packet age. Contacts are cropped to fingertip pads and transformed into sensor coordinates. Object, table and floor partners are included, without exposing partner identity.

The 25-value signal is a physics-based tactile proxy—not RGB, TacMap or a validated model of capacitive electronics. This run does not change contact materials.

Known fidelity gap: self-collision is OFF; removing the fixed torso proxy does not preserve arm-torso collision pairs.

Online SAPG learns lifting, not yet reposing

Run 547, September 13–14. These are logged training statistics from exploratory environments; no held-out evaluation or baseline-matched ablation was run.

1.233B

cumulative transitions

318.6

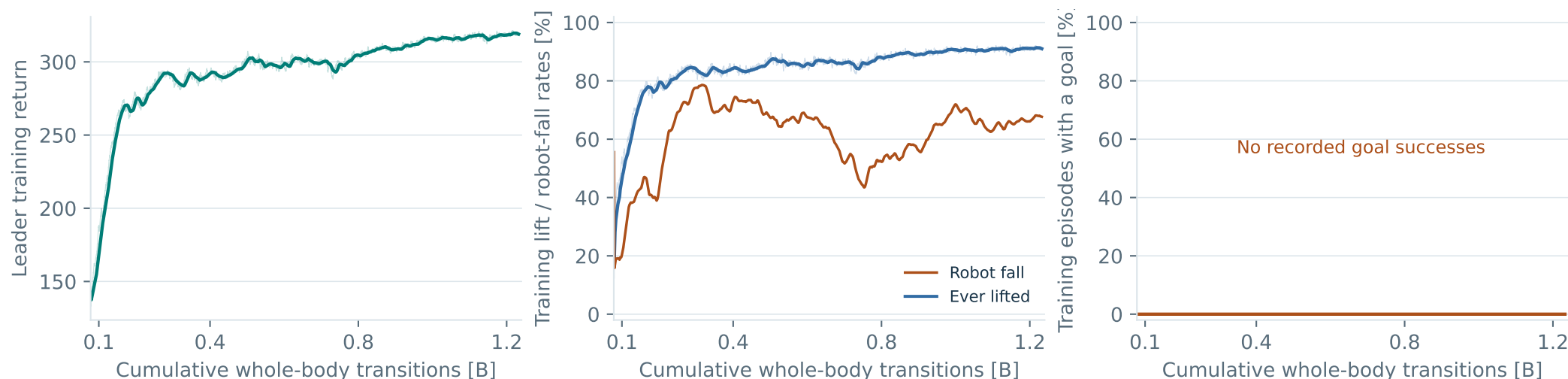
last leader training return

91.0%

last training lift-rate window

0.0%

last training goal-rate window



Raw reward/lift/goal values + trailing 100-update means; robot fall shows the trailing mean. Rates are recent training-episode windows.

Last window: 67% robot falls despite 91% lifting. These events can occur in the same episode; this is not stable reposing.

SAME MANIPULATION OBJECTIVE

Approach progress, lifting, orientation/keypoint progress and goal bonuses; original arm/finger motion penalties. **No new balance reward or online teacher loss.**

Goal success: all four keypoint errors < 0.015 m; 10 accumulated near-goal control steps. Lifting once is not a reposing success. A body fall terminates the episode and loses future task reward.

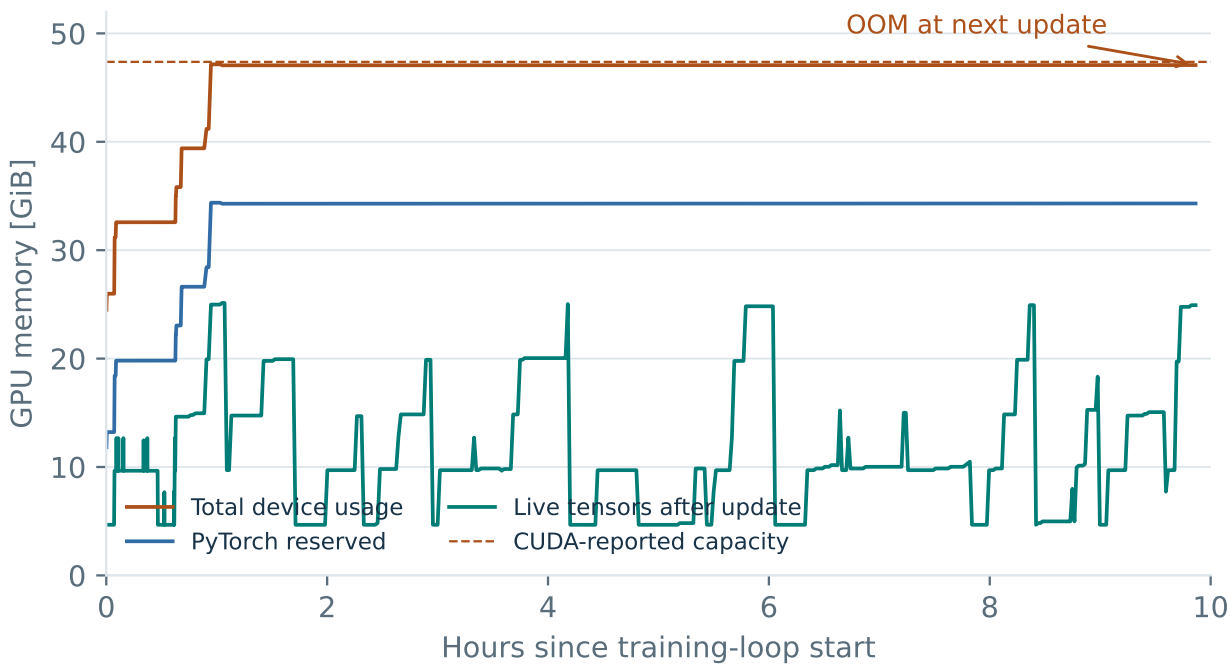
UNCHANGED SAPG TRAINER; NEW ACTOR INTERFACE

6 exploration groups × 1,536 envs; horizon / recurrent sequence 16; minibatch 36,864; two mini-epochs. Actor LR 10^{-5} ; critic LR 10^{-4} ; $\gamma = 0.99$; $\lambda = 0.95$; PPO clip 0.1; mixed precision.

Existing leader/follower experience sharing and learned exploration embeddings are retained. **Best means highest leader training return, not the best evaluated success rate.**

One-GPU training works; long-run memory does not

Job 547 resumed at 78,544,896 transitions and completed 7,826 new updates on one RTX 6000 Ada. It failed at 05:55 Dubai time on September 14, after 9 h 57 min 57 s allocated wall time.



Confirmed failure: observation-batch copy

SAPG torch.cat requested 1.37 GiB.
Only 303.75 MiB free; 31.07 GiB live PyTorch allocations. OOM occurred while repeating observations across exploration groups.

Why the early fit was insufficient: device usage grew from ~26 to ~47 GiB. The failure site is known, but the cause of growing tensor lifetimes / workload peaks is not yet established. This is not proof of a specific leak or fragmentation bug.

Earlier failures: 517 OOM at 12,288 envs; 528 native segfault at 9,216. Periodic stack dumping was disabled as a suspected trigger before 547. No repeat segfault was observed in this run.

Stage-level samples: every update initially, then every 16 updates. “Live” is measured after train_epoch, before outer-loop logging/cleanup. Reserved memory includes live allocations and cache; the device gap includes simulator/runtime allocations.

Evidence	What is verified	What is not verified
49 CPU tests / 4 preflight	Contracts, action/history handling, resume checks	Native stability or learned manipulation success
Ground-contact check 515	Normal-force reconstruction error $\leq 1.91\text{e-}6$ N	Real fingertip force calibration / all contacts
Finite-speed safeguards	Affected env resets above 1,000 rad/s	Numerical cause; 2,385 outliers in run 547

What was delivered—and what remains

Implementation follows the requested direct SONIC fine-tuning route. It is not a reproduction of the full GRAIL video-data pipeline or a completed bimanual locomotion policy.

Proposal item	Status	Implemented mechanism / remaining limit
Reuse the trained SAPG skill	Implemented	Teacher 383 → bootstrap 503 → one 35-action SONIC/SAPG actor
Use the original manipulation task	Implemented	Source reward/goal/DR + BPS-128 + 70 Hz touch; contract checks
Learn whole-body manipulation	Partial	Last training window: 91% lift / 67% robot fall / 0% goal
Walk and stabilize while grasping	Not validated	Legs are controlled; standing reference, no footstep planner/curriculum
Bimanual manipulation	Not implemented	Left fingers are neutral; 41-action extension still needs training/data
Stable one-GPU long training	Unresolved	9,216 envs ran nearly 10 hours, then CUDA OOM; no active job

NEXT WORK, IN ORDER

1. Trace growing SAPG allocations; reduce unnecessary batch copies without changing learning math; test long-run memory with the full optimizer workload.
2. Resume, then validate precise reposing and balance across objects, payloads and perturbations.
3. Add varying approach distances and stepping requirements; then left-hand observations, commands and demonstrations for the bimanual stage.

RECOVERABLE STATE

Last completed: 1,232,535,552 transitions / epoch 8,316.
Latest saved: 1,223,688,192 / epoch 8,256.
Best-return saved: 1,216,462,848 / epoch 8,207.
Both checkpoint archives passed CRC checks. Complete saves include actor and critic optimizers; atomic latest saves occur every 64 updates. Source checkpoints remain unchanged.

AUDIT LINKS · figures use exported raw logs; provenance and plotted CSV/JSON are supplied alongside the PDF

Actor / critic implementation	Environment and body actions	Trainer / source-task checks	W&B run 547
SONIC source	GRAIL reference project	Related diagnostic-crash report	